# Giovanni Sullutrone

LLM safety researcher · refusal, evaluation and agents

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I like finding the places where language models behave in ways nobody planned for: a harmless request refused because of the documents around it, or an agent that senses a question is wrong but cannot say why. Most of my papers started from a discussion with researchers or users, or from a claim in the literature that I wanted to test. Lately, I have started looking for the causes inside the models themselves, with interpretability methods.

## EDUCATION

**PhD in Information and Communication Technologies**, *University of Modena and Reggio Emilia* Nov 2023 – Oct 2026
UNIMORE, DBGroup, Department of Engineering "Enzo Ferrari". Supervisor: Prof. Sonia Bergamaschi. Three-year programme, November 2023 to October 2026. Defence expected in early 2027, degree not yet awarded.

**MSc in Computer Engineering**, *University of Salerno* Oct 2021 – Oct 2023
Artificial Intelligence and Intelligent Robotics track. Final mark 110/110 cum laude (Italian scale, maximum 110).

**BSc in Computer Engineering**, *University of Salerno* Oct 2018 – Oct 2021
Final mark 110/110 (Italian scale, maximum 110).

## RESEARCH EXPERIENCE

**PhD Candidate (research focus: LLM safety and evaluation)**, *UNIMORE, Modena, Italy* Nov 2023 – Oct 2026

- Led three first-author studies of when models refuse or ask for clarification: COVER in retrieval-augmented generation (RAG), Text-to-Refused-SQL in Text-to-SQL over databases with personal data, and ABISS in multi-turn conversations with Text-to-SQL agents.
- Built the evaluation pipelines for these three studies, running open models with vLLM on A100 GPU nodes alongside the OpenAI and Gemini APIs. Together they tested 21 models from 3B to 122B parameters, among them Llama-3-70B, GPT-OSS-120B, GPT-4o-mini, Gemini 1.5 Flash and Gemini 2.0 Flash-Lite, on question answering, translation, summarisation and SQL generation.
- Built the scoring stages: Llama-Guard-3-8B flags unsafe passages, and an LLM judge (Mistral-Small-2501) labels refusals. Validated the judge against human labels: two co-authors independently labelled 100 sampled answers, and their labels matched the judge's in all 100 cases.

**Visiting Researcher**, *DARE Lab, Athena Research Center, Athens, Greece* (host: Dr Georgia Koutrika) Apr 2025 – Jul 2025

- Designed ABISS, a benchmark that evaluates Text-to-SQL agents on ambiguous and unanswerable requests through multi-turn interaction with simulated users. Three 120B-class open models (GPT-OSS-120B, Qwen3.5-122B-A10B, Nemotron-3-Super-120B) generate and validate the questions and act as the simulated user.

**Machine Learning Researcher (research grant)**, *A.I. Tech S.r.l., Fisciano, Italy* 2023

- Defined the user requirements, then designed and implemented a monitoring and prediction system for public transport management.

## SELECTED FIRST-AUTHOR PUBLICATIONS

**COVER: Context-Driven Over-Refusal Verification in LLMs** Findings of ACL 2025
**G. Sullutrone**, R. A. Vigliermo, S. Bergamaschi, L. Sala paper code

Shows that the documents given to a model can make it refuse a harmless task, which the paper calls context-driven over-refusal. The study tested 13 models on 8 tasks and 3 system prompts over the same retrieved documents, with five samples per prompt. Translation and summarisation were refused more often than question answering (20% and 18% against 11% on average), and some model and task settings refused every request. GPT-4o-mini refused 78% of hadith translations under an ethical system prompt, and a chain-of-thought instruction raised Llama-3-8B's refusals on question answering from 25% to 43% under a helpful one. Later preprints cite COVER, among them work that turns this kind of refusal into blocking attacks on RAG systems.

**Text-to-Refused-SQL: A Comprehensive Evaluation of LLMs Refusal in Text-to-SQL** SEBD 2025 (Italy)
**G. Sullutrone**, L. Sala, L. Trigiante, S. Bergamaschi paper

Shows that personal and sensitive fields in a database schema can make models refuse ordinary questions, which the paper calls schema-driven over-refusal. The study tested 7 models on 319 healthcare databases augmented with synthetic personal and sensitive fields, with five samples per prompt. Under a strict ethical system prompt, Llama-2 refused 58% of questions that used no personal field, while GPT-4o-mini, Gemini 2.0 Flash-Lite and Phi-4 stayed below 1% in every setting. Telling the model that access to the personal data was authorised raised refusal almost everywhere a model refused at all, often more than doubling it.

**ABISS: Evaluating Text-to-SQL Systems Through Agent Interaction** Findings of EACL 2027
**G. Sullutrone**, L. Sala, S. Aftar, G. Koutrika, S. Bergamaschi paper
Sorts the ways a database question can be ambiguous or impossible to answer into a taxonomy of 8 categories, and builds 7,145 such questions on BIRD and Spider with a multi-agent pipeline of three open models, ten validation stages and majority votes. Two experts judged the category correct in 92% of ratings on 52 sampled questions. In conversations with simulated users, every agent recognised a problematic question more often than it could name the right category (best model on BIRD: 70.5% against 59.1%). Even with the correct category supplied, execution accuracy on ambiguous questions averaged only 32% on BIRD and 40% on Spider.

**Sensitive Topics Retrieval in Digital Libraries: A Case Study of ḥadīṯ collections** TPDL 2024
**G. Sullutrone**, R. A. Vigliermo, L. Sala, S. Bergamaschi paper
Asks whether search tools make sensitive passages of a ḥadīṯ collection harder to find, using ten embedding models and about 12,000 questions generated from roughly 4,000 passages. Sensitive passages were usually retrieved but ranked lower than non-sensitive ones, by up to 26%, while the BM25 baseline varied less.

## OTHER PUBLICATIONS

Eight further works (2024–2026), seven of them co-authored, on Text-to-SQL with LLMs, LLMs in digital humanities, Arabic-script OCR datasets and digital-library platforms, including a JCDL 2024 short paper that won second place for Best Short Paper. Full list: giosullutrone.github.io.

## SOFTWARE AND DATA

**COVER**, the open-source pipeline behind the paper. It runs open models with vLLM and API models from OpenAI and Gemini on the same documents, and scores their answers with Llama Guard and an LLM judge 2025
github.com/giosullutrone/COVER

## OTHER EXPERIENCE

**Software Development Consultant (freelance)**, *DataRiver* Mar 2026 – Jul 2026
Built proofs of concept for clinical and healthcare data collection, from the requirements and planning to the graphical interface and its first use, with testing and technical documentation.

## SKILLS

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| **Research** | LLM safety evaluation, refusal and over-refusal, controlled prompt conditions, LLM judges with human validation, safety classifiers (Llama Guard 3), multi-turn agent evaluation with simulated users, benchmark design, retrieval-augmented generation, Text-to-SQL |
| **Evaluation infrastructure** | vLLM batch inference with tensor parallelism, OpenAI and Gemini APIs, Llama Guard and LLM-judge classifiers, human agreement checks, A100 GPU clusters |
| **Programming** | Python, PyTorch, Hugging Face Transformers, TRL, LoRA fine-tuning, Docker, SQL, Git |

All twelve publications with links: giosullutrone.github.io · Updated November 2026